Sahaj Bhadja

(213) 284-4651 sahaj.bhadja@gmail.com sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Full-stack Software Engineer with 2+ years building **Python** FastAPI backends and **React** TypeScript frontends for AI platforms, integrating **LLM APIs**, deploying with Docker and CI/CD on Azure and AWS, and enforcing secure access.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Advanced NLP, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Data Retrieval, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Restructured fault-tolerant asynchronous **LLM orchestration** with FastAPI background workers, explicit job states, and failure isolation, processing 1K+ AI jobs at 3x throughput for an education platform without cascading failures.
- Designed multi-tenant storage on **Azure** SQL and Blob Storage with **row-level security** protecting sensitive data for 500+ users, and shipped integrations through CI/CD at 99%+ reliability across 5K+ monthly jobs.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built a full-stack platform with a **React** frontend and Python Flask backend, owning upload, rubric, and grading features from development to production for 150+ instructors and 10K+ daily requests.
- Integrated **OpenAI** and Anthropic models plus Qwen-VL behind a switchable provider layer to automate grading and feedback, with authenticated sessions and per-call cost tracking, cutting turnaround 60%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Translated stakeholder requirements from Microsoft and GitHub teams into working features in Agile sprints, building **automations** that cut manual data entry 35% and API response times 30%.
- Wrote 20+ automated **unit and integration tests** with PyTest and Mockito in nightly Jenkins pipelines across Windows, Linux, and macOS, publishing coverage reports and failure logs for the whole team.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a **TypeScript** React platform with REST microservices, JWT authentication, and role-based access for customer, billing, and lead workflows, syncing 2M+ records across multiple locations in real time.
- Deployed **Docker** images to Azure App Services through Jenkins CI/CD with staged releases, and monitored production with Sentry, Azure Monitor, and Kibana, cutting deployment time 50% for every production rollout.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Gathered requirements with the client, mapped 12+ integration points, and presented the architecture, then built Flask APIs and a React TypeScript dashboard with responses under 200ms across 50K+ records.
- Implemented **secure authentication** with JWT and wallet-signature login, and automated refund handling through idempotent payment webhooks, cutting manual refund work about 70% across 50K+ booking and payment records.

Skills

Languages: Python, TypeScript, JavaScript, SQL, Java
Full Stack: FastAPI, Flask, React, Next.js, Node.js, REST APIs, Async Processing, Code Reviews, Technical Documentation
AI: LLM APIs, OpenAI, Claude, Prompt Engineering, RAG, Embeddings, FAISS, Linux
Cloud & DevOps: Azure, Azure SQL, Azure App Services, AWS, Docker, Kubernetes, CI/CD, GitHub Actions, Jenkins
Security & Quality: JWT, Role-Based Access, Row-Level Security, PyTest, Jest, Sentry, Azure Monitor, Git, Agile

Projects

Internal Knowledge Assistant with SSO | *FastAPI, React, TypeScript, Azure OpenAI, OIDC*

- Designed an internal assistant that answers questions from company documents with citations, signs users in through **OIDC** single sign-on, and filters results to what each user's **role** may access.

Workflow Automation Agent | *Python, LLM Tool Calling, FastAPI, OpenTelemetry*

- Designed a **tool-calling** agent that drafts routine business requests, fills forms through internal APIs after user approval, and logs each step with traces for **monitoring** and audit.

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, Clerk, Convex*

- Built a secure **document** Q&A platform with Clerk authentication, semantic search, and summarization over 50K+ documents, reaching 95% query accuracy and cutting review time 60% for 200+ users.